Session 8 – AADK Task 8

UI Testing and Art Navigation

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Session: 8
Task Title: UI Testing and Art Navigation

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1. Introduction

This session focuses on understanding professional testing tools used in Jetpack Compose applications. The objective is to simulate user behavior, validate UI output, and ensure that state changes correctly reflect in the UI.

In this task, a console-based Art Gallery Navigator was implemented to practice navigation logic, state handling, and testable module design before integrating it into a Compose UI.

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2. Aim

To develop an Art Gallery Navigator that:

- Displays artworks with title, artist, and year
- Allows navigation using Next and Previous logic
- Implements wrap-around navigation
- Supports a Favourite toggle feature
- Separates business logic for testability

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3. Objectives

- Implement state management using index tracking
- Apply modular arithmetic for wrap logic
- Design testable business logic
- Simulate unit testing for navigation behaviour
- Understand the relationship between state and UI

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4. System Design

4.1 Data Model

A data class Artwork was created to store:

- Title
- Artist
- Year

4.2 Business Logic Layer

An ArtGalleryManager class manages:

- Current artwork index
- Navigation (Next / Previous)
- Favourite toggle
- List of favourite artworks

This separation ensures that the business logic is independent of UI, making it easier to test.

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5. State Variables Used

State Variable	Type	Purpose
currentIndex	Int	Tracks current artwork
favourites	MutableList<Artwork>	Stores favourite artworks

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6. Navigation Logic

Next Logic

```
currentIndex = (currentIndex + 1) % artworks.size
```

This ensures wrap-around when the last artwork is reached.

Previous Logic

If index becomes negative, it wraps to the last artwork.

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7. Test Strategy

The application includes simulated unit test functions:

Test Case	What is Verified	Expected Result
Next Wrap Logic	Moving beyond last item	Index resets to 0
Previous Wrap Logic	Moving before first item	Index moves to last
Favourite Toggle	Add/Remove favourite	List updates correctly

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8. Sample Output

```
----- ART GALLERY NAVIGATOR -----
Viewing: Starry Night (Vincent van Gogh, 1889)
-> Next Artwork
Viewing: Mona Lisa (Leonardo da Vinci, 1503)
-> Next Artwork
Viewing: The Persistence of Memory (Salvador Dali, 1931)
-> Previous Artwork
Viewing: Mona Lisa (Leonardo da Vinci, 1503)
-> Toggle Favourite Favourites List:
Mona Lisa by Leonardo da Vinci

----- RUNNING TEST CASES -----
PASS: Next wrap logic working
PASS: Previous wrap logic working
PASS: Favourite added
PASS: Favourite removed
```

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9. Learning Outcomes

Through this task, I learned:

- How to design modular and testable code
- Importance of separating business logic from UI
- How wrap logic prevents index errors
- How state changes must be verified through testing
- Basic simulation of unit test behaviour

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10. Reflection

Planning tests alongside implementation improved code clarity and reliability. Separating logic from UI makes debugging easier and reduces tight coupling.

One challenge is ensuring that state changes are consistently reflected in UI during real Compose testing. Proper structuring and use of testing tools like ComposeTestRule will help mitigate this issue.
Testing early helps identify edge cases such as wrap conditions and state resets before deployment.

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11. Conclusion

The Art Gallery Navigator was successfully implemented with navigation, state management, and testable business logic. This task provided foundational understanding of UI testing principles and prepared the application structure for integration with Jetpack Compose UI testing frameworks.